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1.1 Overview

This diagram describes the full object-oriented class model of the 3Dex platform. It defines each domain entity as a class with attributes and operations, then maps the inheritance hierarchy and associations among them.

1.2 Inheritance Hierarchy

All user roles extend a single base **User** class:

- **Customer** extends **User** - can shop, checkout, review, and write posts.
- **Artist** extends **User** - can upload and manage 3D models.
- **Provider** extends **User** - can manage print jobs.
- **Admin** extends **User** - can moderate content and users.

This matches the Prisma **Role** enum (**CUSTOMER**, **ARTIST**, **ADMIN**, **PROVIDER**) stored on the single **User** table.

1.3 Classes

1.3.1 User

The base entity for all platform actors.

Field	Type	Description
id	String	UUID primary key
email	String	Unique login email
username	String	Unique handle
password	String	bcrypt-hashed; null for OAuth-only accounts
google_id	String	Unique Google OAuth subject ID
role	Role	One of CUSTOMER, ARTIST, PROVIDER, ADMIN
account_status	Account_Status	PENDING, APPROVED, REJECTED
status_history	Json[]	Audit trail of status changes
approved_at	DateTime	Timestamp of approval
rejected_at	DateTime	Timestamp of rejection
display_name	String	Public display name
bio	String	Profile bio
avatar_url	String	S3 key for avatar image
banner_url	String	S3 key for profile banner
location	String	Free-text location
website	String	Personal website link
phone_number	String	Contact number
addresses	Json[]	Shipping addresses array
portfolio	Json[]	Portfolio items for Artists/Providers
provider_config	Json	Print capabilities config for Providers
social_twitter	String	Twitter/X handle
social_instagram	String	Instagram handle
social_artstation	String	ArtStation handle
social_behance	String	Behance handle
rating	Float	Aggregate rating from User_Review records
review_count	Int	Total number of reviews received
two_factor_enabled	Boolean	2FA preference flag
last_login_at	DateTime	Timestamp of most recent login

Field	Type	Description
show_nsfw	Boolean	NSFW content visibility preference
dexie_enabled	Boolean	Whether Dexie AI assistant is active
created_at	DateTime	Account creation timestamp
updated_at	DateTime	Last modification timestamp

Operations: `register()`, `login()`, `updateProfile()`, `changePassword()`, `deleteAccount()`

1.3.2 Customer (extends User)

A base user who can browse, buy, and engage with content.

Additional Operations: `addToCart(model_id, qty)`, `checkout()`, `purchaseModel(model_id)`, `writeReview(model_id, rating)`

1.3.3 Artist (extends User)

A creator who uploads and sells 3D models. Requires admin approval (`account_status = APPROVED`).

Additional Operations: `uploadModel(data)`, `updateModel(id, data)`, `deleteModel(id)`, `viewSalesStats()`

1.3.4 Provider (extends User)

A 3D printing service provider. Also requires admin approval.

Additional Operations: `acceptPrintJob(order_id)`, `updatePrintStatus(order_id, status)`, `updateProviderConfig(config)`

1.3.5 Admin (extends User)

A platform moderator with elevated privileges.

Additional Operations: `approveModel(id)`, `rejectModel(id)`, `approveUser(id)`, `rejectUser(id)`, `suspendUser(id)`, `generatePlatformReport()`, `viewAuditLogs()`

1.3.6 Model

A 3D asset listed on the platform.

Field	Type	Description
id	String	UUID primary key
title	String	Model name
description	String	Optional description
price	Int	Price in IDR; 0 for free
file_url	String	S3 key for the GLB/GLTF file
preview_url	String	S3 key for WebP preview image
gallery_urls	String[]	S3 keys for additional gallery images
status	Model_Status	PENDING, APPROVED, REJECTED

Field	Type	Description
is_nsfw	Boolean	NSFW content flag
license	License_Type	PERSONAL_USE or COMMERCIAL_USE
is_printable	Boolean	Whether the model supports print orders
file_format	String	Default “glb”
embedding	vector	384-dimension semantic embedding (pgvector)
artist_id	String	FK to User (Artist role)
category_id	String	FK to Category
avg_rating	Float	Aggregated average rating
review_count	Int	Number of reviews
created_at	DateTime	Creation timestamp

Operations: `updateStatus()`, `calculateAvgRating()`, `incrementReviewCount()`

1.3.7 Purchase

A permanent record of a user owning a model after payment.

Field	Type	Description
id	String	UUID primary key
user_id	String	FK to User
model_id	String	FK to Model (nullable for deleted models)
price_paid	Int	Price at time of purchase in IDR
license	License_Type	License type at time of purchase
created_at	DateTime	Purchase timestamp

Constraint: Unique on (`user_id`, `model_id`).

1.3.8 Category

A taxonomy grouping for models.

Field	Type	Notes
id	String	UUID primary key
name	String	Unique display name
slug	String	Unique URL slug
created_at	DateTime	Creation timestamp

1.3.9 Tag

A free-form label attached to models. Many-to-many with `Model`.

Field	Type	Notes
id	String	UUID primary key
name	String	Unique tag name

Field	Type	Notes
created_at	DateTime	Creation timestamp

1.3.10 Review

A model review written by a verified purchaser.

Field	Type	Notes
id	String	UUID primary key
rating	Int	1-5 stars
comment	String	Optional text
user_id	String	FK to User
model_id	String	FK to Model
created_at	DateTime	Creation timestamp
updated_at	DateTime	Last edit timestamp

Constraint: Unique on (user_id, model_id).

1.3.11 Wishlist

A saved model reference per user. Toggle-based.

Field	Type
id	String
user_id	String
model_id	String
created_at	DateTime

Constraint: Unique on (user_id, model_id).

1.3.12 Cart_Item

A staging area before checkout. All digital cart items have quantity 1.

Field	Type
id	String
user_id	String
model_id	String
quantity	Int
created_at	DateTime
updated_at	DateTime

Constraint: Unique on (user_id, model_id).

1.3.13 Order

Represents a checkout transaction, either for digital assets or print jobs.

Field	Type	Description
id	String	UUID primary key
user_id	String	FK to User (customer)
provider_id	String	FK to User (Provider); null for ASSET orders
total_amount	Int	Total in IDR
status	Order_Status	PENDING, PAID, FAILED, CANCELLED
type	Order_Type	ASSET or PRINT_JOB
courier_name	String	Courier for print jobs
tracking_number	String	Tracking code for shipped print jobs
shipping_address	Json	Address JSON for print delivery
proof_urls	String[]	Photo evidence for shipped jobs
snap_token	String	Midtrans Snap token
snap_redirect_url	String	Midtrans redirect URL
created_at	DateTime	Creation timestamp
updated_at	DateTime	Last modification timestamp

1.3.14 Order_Item

A single line within an order referencing one model.

Field	Type	Description
id	String	UUID primary key
order_id	String	FK to Order
model_id	String	FK to Model (nullable if model deleted)
price	Int	Unit price in IDR at time of purchase
quantity	Int	Quantity (1 for digital, configurable for print)
print_config	Json	Material, infill, color, scale settings
print_status	Print_Status	PENDING, ACCEPTED, PROCESSING, SHIPPED, DELIVERED, CANCELLED
created_at	DateTime	Creation timestamp

1.3.15 Payment

A Midtrans transaction record linked to an order.

Field	Type	Description
id	String	UUID primary key
order_id	String	FK to Order
transaction_id	String	Unique Midtrans transaction ID

Field	Type	Description
payment_type	String	e.g. “credit_card”, “gopay”
gross_amount	Int	Total charged in IDR
transaction_status	String	Midtrans status string
fraud_status	String	Midtrans fraud detection result
raw_response	Json	Full Midtrans webhook payload
created_at	DateTime	Webhook received timestamp

1.3.16 User_Review

A review of an Artist or Provider by a verified transacting user.

Field	Type	Notes
id	String	UUID primary key
reviewer_id	String	FK to User (reviewer)
target_user_id	String	FK to User (artist or provider)
rating	Int	1-5 stars
comment	String	Optional comment
created_at	DateTime	Creation timestamp

Constraint: Unique on (reviewer_id, target_user_id).

1.3.17 Post

A community post created by Artists or Providers.

Field	Type	Notes
id	String	UUID primary key
user_id	String	FK to User
caption	String	Post text
media_urls	String[]	S3 keys for uploaded images
is_nsfw	Boolean	NSFW flag
like_count	Int	Denormalized count
comment_count	Int	Denormalized count
created_at	DateTime	Creation timestamp
updated_at	DateTime	Last edit timestamp

1.3.18 Post_Like

Join record between a User and a Post for the like feature.

Field	Type
id	String
user_id	String
post_id	String

Field	Type
created_at	DateTime

Constraint: Unique on (user_id, post_id).

1.3.19 Post_Comment

A text comment attached to a Post.

Field	Type
id	String
user_id	String
post_id	String
content	String
created_at	DateTime
updated_at	DateTime

1.3.20 Stats

A periodic aggregation record created by the admin dashboard's aggregation trigger.

Field	Type	Description
id	String	UUID primary key
period_start	DateTime	Aggregation window start
period_end	DateTime	Aggregation window end
data	Json	Aggregated metrics JSON
created_at	DateTime	When this record was computed

1.3.21 Follow

A directional social connection from one user to another.

Field	Type
id	String
follower_id	String
following_id	String
created_at	DateTime

Constraint: Unique on (follower_id, following_id).

1.3.22 Notification

An in-app notification record delivered to a user.

Field	Type	Description
id	String	UUID primary key
user_id	String	FK to recipient User
type	String	Notification category string
title	String	Short heading
message	String	Body text
is_read	Boolean	Read state
data	Json	Optional metadata payload
created_at	DateTime	Delivery timestamp

1.3.23 Collection

A user-curated named group of models.

Field	Type
id	String
user_id	String
name	String
description	String
is_public	Boolean
created_at	DateTime
updated_at	DateTime

1.3.24 Collection_Item

A join record linking a Collection to a Model.

Field	Type
id	String
collection_id	String
model_id	String
added_at	DateTime

Constraint: Unique on (collection_id, model_id).

1.3.25 Report

A user-submitted abuse report targeting a model, post, or comment.

Field	Type	Description
id	String	UUID primary key

Field	Type	Description
reporter_id	String	FK to User who filed the report
target_type	Report_Target	MODEL, POST, or COMMENT
model_id	String	FK to Model (optional)
post_id	String	FK to Post (optional)
comment_id	String	FK to Post_Comment (optional)
reason	String	Free-text reason
status	Report_Status	PENDING, REVIEWED, DISMISSED
created_at	DateTime	Submission timestamp

1.3.26 Admin_Audit_Log

An immutable record of every admin action taken on the platform.

Field	Type	Description
id	String	UUID primary key
admin_id	String	FK to User (Admin role)
action	Audit_Action	DELETE_MODEL, DELETE_POST, DELETE_COMMENT, BAN_USER, REJECT_MODEL, REJECT_USER, APPROVE_USER, USER_STEP_DOWN
target_id	String	ID of the affected entity
target_type	String	Human-readable entity type string
reason	String	Mandatory justification
metadata	Json	Extra context (title, username, price, etc.)
created_at	DateTime	Action timestamp

1.4 Associations Summary

From	Relationship	To	Notes
User	owns (composition)	Cart_Item, Wishlist, Post	Deleted when user deleted
Order	contains (composition)	Order_Item, Payment	Part of order
Post	has (composition)	Post_Like, Post_Comment	Part of post
Collection	contains (composition)	Collection_Item	Part of collection
Artist	creates	Model	One Artist, many Models
Category	categorizes	Model	Optional FK

2.2 Entities

2.2.1 User

The single unified table for all platform actors (Customer, Artist, Provider, Admin).

Column	Type	Constraint	Notes
id	String	PK	UUID
email	String	UK	Unique login email
username	String	UK	Unique handle
password	String		Nullable for OAuth-only accounts
google_id	String	UK	Google OAuth subject
role	Role		CUSTOMER, ARTIST, PROVIDER, ADMIN
account_status	Account_Status		PENDING, APPROVED, REJECTED
status_history	Json[]		Array of status change records
approved_at	DateTime		
rejected_at	DateTime		
display_name	String		
bio	String		
avatar_url	String		S3 key
banner_url	String		S3 key
location	String		
website	String		
phone_number	String		
addresses	Json[]		Shipping address array
portfolio	Json[]		Artist/Provider portfolio
provider_config	Json		Provider print capabilities
social_twitter	String		
social_instagram	String		
social_artstation	String		
social_behance	String		
rating	Float		Aggregated score
review_count	Int		Denormalized count
two_factor_enabled	Boolean		
last_login_at	DateTime		
show_nsfw	Boolean		Content filter preference
dexie_enabled	Boolean		Dexie AI toggle
created_at	DateTime	NOT NULL	
updated_at	DateTime	NOT NULL	Auto-managed

2.2.2 Model

A 3D asset uploaded by an Artist.

Column	Type	Constraint	Notes
id	String	PK	UUID
title	String	NOT NULL	
description	String		
price	Int	NOT NULL	IDR; 0 = free
file_url	String	NOT NULL	S3 key for GLB/GLTF
preview_url	String		S3 key for WebP preview

Column	Type	Constraint	Notes
gallery_urls	String[]		Additional image keys
status	Model_Status		PENDING, APPROVED, REJECTED
is_nsfw	Boolean		
license	License_Type		PERSONAL_USE, COMMERCIAL_USE
is_printable	Boolean		
file_format	String		Default “glb”
embedding	vector		384-dim pgvector for semantic search
artist_id	String	FK -> User	Required
category_id	String	FK -> Category	Optional
avg_rating	Float		Aggregated rating
review_count	Int		Denormalized count
created_at	DateTime	NOT NULL	

2.2.3 Purchase

Immutable record created when payment is confirmed for a digital model.

Column	Type	Constraint
id	String	PK
user_id	String	FK -> User
model_id	String	FK -> Model (nullable)
price_paid	Int	NOT NULL
license	License_Type	NOT NULL
created_at	DateTime	NOT NULL

Unique: (user_id, model_id)

2.2.4 Category

A top-level taxonomy for models.

Column	Type	Constraint
id	String	PK
name	String	UK
slug	String	UK
created_at	DateTime	NOT NULL

2.2.5 Tag

A label for filtering and semantic grouping.

Column	Type	Constraint
id	String	PK
name	String	UK

Column	Type	Constraint
created_at	DateTime	NOT NULL

2.2.6 Review

A star rating and optional comment left by a purchaser.

Column	Type	Constraint
id	String	PK
rating	Int	NOT NULL
comment	String	
user_id	String	FK -> User
model_id	String	FK -> Model
created_at	DateTime	NOT NULL
updated_at	DateTime	NOT NULL

Unique: (user_id, model_id)

2.2.7 Wishlist

A saved reference to a model.

Column	Type	Constraint
id	String	PK
user_id	String	FK -> User
model_id	String	FK -> Model
created_at	DateTime	NOT NULL

Unique: (user_id, model_id)

2.2.8 Cart_Item

A model staged for checkout.

Column	Type	Constraint
id	String	PK
user_id	String	FK -> User
model_id	String	FK -> Model
quantity	Int	NOT NULL
created_at	DateTime	NOT NULL
updated_at	DateTime	NOT NULL

Unique: (user_id, model_id)

2.2.9 Order

An order for digital assets or a 3D print job.

Column	Type	Constraint	Notes
id	String	PK	UUID
user_id	String	FK -> User	Buyer
provider_id	String	FK -> User	Provider; null for ASSET orders
total_amount	Int	NOT NULL	IDR
status	Order_Status	NOT NULL	PENDING, PAID, FAILED, CANCELLED
type	Order_Type	NOT NULL	ASSET or PRINT_JOB
courier_name	String		Print orders only
tracking_number	String		Print orders only
shipping_address	Json		Print orders only
proof_urls	String[]		Photo proofs for shipped jobs
snap_token	String		Midtrans token
snap_redirect_url	String		Midtrans redirect
created_at	DateTime	NOT NULL	
updated_at	DateTime	NOT NULL	

2.2.10 Order_Item

A single line item within an order.

Column	Type	Constraint	Notes
id	String	PK	UUID
order_id	String	FK -> Order	
model_id	String	FK -> Model	Nullable if model deleted
price	Int	NOT NULL	IDR at time of purchase
quantity	Int	NOT NULL	
print_config	Json		Material, color, infill settings
print_status	Print_Status	NOT NULL	PENDING through DELIVERED
created_at	DateTime	NOT NULL	

2.2.11 Payment

A Midtrans transaction record.

Column	Type	Constraint
id	String	PK
order_id	String	FK -> Order
transaction_id	String	UK
payment_type	String	NOT NULL
gross_amount	Int	NOT NULL
transaction_status	String	NOT NULL
fraud_status	String	
raw_response	Json	
created_at	DateTime	NOT NULL

2.2.12 User_Review

A rating of an Artist or Provider by a verified customer.

Column	Type	Constraint
id	String	PK
reviewer_id	String	FK -> User
target_user_id	String	FK -> User
rating	Int	NOT NULL
comment	String	
created_at	DateTime	NOT NULL

Unique: (reviewer_id, target_user_id)

2.2.13 Post

A community post with images and an optional caption.

Column	Type	Constraint
id	String	PK
user_id	String	FK -> User
caption	String	
media_urls	String[]	
is_nsfw	Boolean	
like_count	Int	NOT NULL
comment_count	Int	NOT NULL
created_at	DateTime	NOT NULL
updated_at	DateTime	NOT NULL

2.2.14 Post_Like

Join table for the like relationship.

Column	Type	Constraint
id	String	PK
user_id	String	FK -> User
post_id	String	FK -> Post
created_at	DateTime	NOT NULL

Unique: (user_id, post_id)

2.2.15 Post_Comment

A text comment on a post.

Column	Type	Constraint
id	String	PK
user_id	String	FK -> User
post_id	String	FK -> Post
content	String	NOT NULL
created_at	DateTime	NOT NULL
updated_at	DateTime	NOT NULL

2.2.16 Stats

A point-in-time aggregation snapshot.

Column	Type	Constraint
id	String	PK
period_start	DateTime	NOT NULL
period_end	DateTime	NOT NULL
data	Json	NOT NULL
created_at	DateTime	NOT NULL

2.2.17 Follow

A directed edge in the social graph.

Column	Type	Constraint
id	String	PK
follower_id	String	FK -> User
following_id	String	FK -> User
created_at	DateTime	NOT NULL

Unique: (follower_id, following_id)

2.2.18 Notification

An in-app notification pushed to a user.

Column	Type	Constraint
id	String	PK
user_id	String	FK -> User
type	String	NOT NULL
title	String	NOT NULL
message	String	NOT NULL
is_read	Boolean	NOT NULL
data	Json	
created_at	DateTime	NOT NULL

2.2.19 Collection

A named list of models curated by a user.

Column	Type	Constraint
id	String	PK
user_id	String	FK -> User
name	String	NOT NULL
description	String	
is_public	Boolean	NOT NULL
created_at	DateTime	NOT NULL
updated_at	DateTime	NOT NULL

2.2.20 Collection_Item

Many-to-many join between Collection and Model.

Column	Type	Constraint
id	String	PK
collection_id	String	FK -> Collection
model_id	String	FK -> Model
added_at	DateTime	NOT NULL

Unique: (collection_id, model_id)

2.2.21 Report

An abuse report submitted by a user.

Column	Type	Constraint
id	String	PK
reporter_id	String	FK -> User
target_type	Report_Target	NOT NULL
model_id	String	FK -> Model (optional)
post_id	String	FK -> Post (optional)
comment_id	String	FK -> Post_Comment (optional)
reason	String	NOT NULL
status	Report_Status	NOT NULL
created_at	DateTime	NOT NULL

2.2.22 Admin_Audit_Log

An append-only record of every admin action.

Column	Type	Constraint
id	String	PK
admin_id	String	FK -> User
action	Audit_Action	NOT NULL
target_id	String	
target_type	String	
reason	String	
metadata	Json	
created_at	DateTime	NOT NULL

2.3 Relationship Summary

Table A	Cardinality	Table B	Via
User	1 to many	Model	artist_id
User	1 to many	Purchase	user_id
User	1 to many	Review	user_id
User	1 to many	Wishlist	user_id
User	1 to many	Cart_Item	user_id
User	1 to many	Order	user_id (customer)
User	1 to many	Order	provider_id (provider)
User	many to many	User_Review	reviewer_id and target_user_id
User	1 to many	Post	user_id
User	1 to many	Post_Like	user_id
User	1 to many	Post_Comment	user_id
User	many to many	Follow	follower_id and following_id
User	1 to many	Notification	user_id
User	1 to many	Collection	user_id
User	1 to many	Report	reporter_id
User	1 to many	Admin_Audit_Log	admin_id
Model	many to 1	Category	category_id
Model	many to many	Tag	implicit join table
Model	1 to many	Purchase	model_id
Model	1 to many	Review	model_id
Model	1 to many	Wishlist	model_id
Model	1 to many	Cart_Item	model_id
Model	1 to many	Order_Item	model_id
Model	1 to many	Collection_Item	model_id
Order	1 to many	Order_Item	order_id
Order	1 to many	Payment	order_id
Post	1 to many	Post_Like	post_id
Post	1 to many	Post_Comment	post_id
Collection	1 to many	Collection_Item	collection_id

3 Use Case Diagram

UML Type: Use Case Diagram **Source:** docs/diagrams/3dex-ucd.drawio.xml

3.1 Overview

This use case diagram represents the comprehensive behavioral model of the 3Dex platform. It maps the complete range of system capabilities to human and external system actors. The diagram separates functions into logical sub-domains and establishes precise <<include>> and <<extend>> relationships to represent execution logic.

3.2 Actors

The 3Dex platform categorizes actors into human participants (inheriting base capabilities in a structured hierarchy) and external system components.

3.2.1 Human Actors

Actor	Description
Guest	An unauthenticated, anonymous visitor. Can perform base browsing activities like searching and viewing.
User/Customer	A registered, authenticated user. Serves as the base role for all active human participants. Inherits all capabilities of Guest .
Artist	A registered creator authorized to upload, manage, and sell 3D models. Inherits all capabilities of User/Customer .
Provider	A registered 3D printing service provider who handles local fabrication and delivery of print orders. Inherits all capabilities of User/Customer .
Admin	The system administrator who moderates content, processes applications, manages system health, and audits actions. Inherits all capabilities of User/Customer .

3.2.2 External & System Actors

Actor	Type	Description
> Midtrans	External Payment Gateway	Responsible for secure transaction processing and payment status webhooks.
> Dexie AI	Gemini-powered Subsystem	Handles intelligent tagline generation and personalized recommendation queries.
> Delivery service	Logistics Gateway	Receives and updates tracking information for physical 3D print products.
> Time	Chron Scheduler	Automated timer that triggers recurring background operations (e.g., weekly best models update).

3.3 Use Cases by Domain

3.3.1 Identity & Profile Management

- **Register and authenticate:** Accessible by **Guest** to establish a new account.
- **Login via OAuth/local:** Included in **Register and authenticate** to authenticate users via local credentials or Google OAuth.
- **Manage user profile:** Central use case for modifying personal information.
 - *Extensions:*
 - * **Update shipping address:** Adds physical shipping details to the user profile.
 - * **Enable 2FA security:** Activates two-factor authentication (2FA).
 - * **Sync social platform:** Connects portfolios (Behance, Artstation, Twitter, etc.).
 - * **Toggle content visibility (NFSW):** Sets content filters.
 - * **Toggle dexie on/off:** Persists AI preferences.
 - * **Submit role application:** Allows standard customers to apply to become an Artist or Provider.

3.3.2 Catalog & Discovery

- **Search models & filters:** Allows searching via categories, pricing, and tag filters.
- **View model details:** High-fidelity Three.js-based viewer page.
- **Display contextual tagline:** Dynamically generates routes/headers via > **Dexie AI**. Extends **Toggle dexie on/off**.
- **Generate personalised picks:** Taste-based recommendations using pgvector similarity. Uses > **Dexie AI** and extends **Toggle dexie on/off** and **View model details**.
- **View best model of the week:** Highlighted showcase on the home page.

3.3.3 Social & Feedback

- **Follow/unfollow creator:** Subscribes to creators to receive feed updates.
- **Create social feed post:** Enabled for **Artist** and **Provider** to publish updates.
- **Like/Comment:** Allows liking and commenting on community posts.
- **Rate counterparty:** Submits provider or artist reviews after a completed print job or model purchase.
- **Submit review:** Submits detailed reviews for models.
- **Report content/user:** Submits abuse or moderation reports.

3.3.4 Curation & Shopping

- **Create curated wishlist/collection:** Allows users to save models.
 - *Includes:*
 - * **Modify collection items:** Adding or removing models from specific folders.
- **View cart:** Interactive checkout preparation stage.
- **Proceed to checkout:** Initiates checkout processing.
 - *Includes:*
 - * **Verify licensing & terms:** Ensures valid license type selection (Personal vs. Commercial).
 - *Extensions:*
 - * **Configure print materials:** Configures infill, colors, and scaling for physical 3D print requests.

3.3.5 Financial & Fulfillment

- **Proceed with payment:** Handles order creation and token acquisition via > **Midtrans**.
- **Unlock downloads:** Grants pre-signed S3 download URLs upon verified order completion.
- **Request earnings payout:** Enables **Artist** and **Provider** to withdraw accumulated earnings.
- **Define print capability:** Enabled for **Provider** to manage capabilities (materials, colors, local pricing).

- **Accept/reject incoming jobs:** Managed by **Provider** to claim print jobs.
- **Update tracking info:** Enables **Provider** and **> Delivery service** to record shipping references.
- **View Delivery Status:** Allows users to track physical shipping progress.
- **View Analytics:** Allows **Artist** and **Provider** to track monthly revenues and performance.

3.3.6 Administrative Operations

- **Moderate content & user:** General content and account lifecycle moderation.
- **Resolve abuse reports:** Reviews and closes submitted reports.
- **Suspend malicious nodes:** Suspends offending users or bans specific items.
- **Audit admin actions:** Records all administrative changes to an append-only ledger (**Admin_Audit_Log**).
- **Manage platform stats:** Views system revenue and model performance snapshots.

3.4 Core Relationships

3.4.1 Include Relationships (<<include>>)

Base Use Case	Included Use Case	Rationale
Register and authenticate	Login via OAuth/local	Every new registration session concludes with active authentication.
Create curated wishlist/collection	Modify collection items	Saving a collection implies the mandatory capacity to structure and edit its items.
Upload 3D Models	Define licenses & pricings	All catalog uploads must have designated licensing terms and prices.
Proceed to checkout	Verify licensing & terms	Checking out requires explicit acceptance of the specific license.

3.4.2 Extend Relationships (<<extend>>)

Extension Use Case	Base Use Case	Extension Point / Condition
Update shipping address	Manage user profile	Triggered conditionally when shipping fields are populated.
Enable 2FA security	Manage user profile	Triggered conditionally when user toggles 2FA security.
Sync social platform	Manage user profile	Triggered conditionally when linking external accounts.
Toggle content visibility (NFSW)	Manage user profile	Triggered conditionally when editing sensitive filters.
Toggle dexie on/off	Manage user profile	Triggered conditionally when customising AI preferences.
Submit role application	Manage user profile	Triggered when a customer applies for Artist/Provider status.
Configure print materials	Proceed to checkout	Triggered when the checkout payload contains a physical print job.
Display contextual tagline	Toggle dexie on/off	Active only when the Dexie AI system toggle is set to enabled .

Extension Use Case	Base Use Case	Extension Point / Condition
Generate personalised picks	Toggle dexie on/off	Active only when the Dexie AI system toggle is set to enabled .
Generate personalised picks	View model details	Triggered conditionally when displaying similarity recommendations.

3.4.3 External System Integrations

- **Midtrans Gateway:**
 - Invoked by **Proceed with payment** to handle transaction processing.
 - Triggers **Unlock downloads** upon receiving a verified payment webhook.
- **Dexie AI Gateway:**
 - Triggers **Display contextual tagline** and **Generate personalised picks** to inject Gemini-powered recommendations into active browsing contexts.
- **Delivery Service:**
 - Triggers **Update tracking info** to sync physical courier status back to the order record.
- **Time Scheduler:**
 - Triggers **Update weekly best models** dynamically every calendar interval.